

UDIT PAUL

New York

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Education

University of California Santa Barbara

September 2023

Doctor of Philosophy (Ph.D.), Computer Science

University of Cape Town

June 2018

Master of Science (M.S.), Electrical Engineering

Technical Focus

Expertise in data science and applied machine learning for large-scale systems, with emphasis on anomaly detection, event modeling, cross-layer measurement analysis, and performance benchmarking. Contributions center on developing scalable analytical frameworks that convert noisy measurement data into operational insights and high-impact models for improving system reliability and product performance.

Professional Experience

Research Scientist

July 2024 – Present

Cisco ThousandEyes

New York City, NY

- Lead research initiatives for the Connected Devices organization, designing performance benchmarks, churn detection models, and large-scale event detection algorithms for Cisco's global device ecosystem.
- Develop cross-layer anomaly detection techniques that fuse network-, transport-, and application-layer telemetry to improve event detection accuracy across Cloud and Enterprise Agents.
- Collaborate with product and engineering teams to translate research insights into deployable detection pipelines, driving reliability improvements across customer-facing monitoring systems.

Data Analyst II

August 2023 – June 2024

Ookla

Seattle, Washington

- Principal architect of the Speedtest Coverage Score—a standardized global connectivity metric used by regulators, operators, and media; benchmarked experience for **100M+** users across **150+** countries.
- Designed a novel anomaly detection methodology for heterogeneous datasets (dense and sparse regions), achieving over **90% precision** and reducing false positives in operational workflows.
- Built and productionized an end-to-end data engineering pipeline using **Airflow** and distributed compute systems to operationalize the anomaly detection framework globally.
- Contributed to “Ookla for Good” by creating engagement templates that strengthened collaboration with universities and research institutions.

Graduate Research Assistant

September 2018 – September 2023

University of California Santa Barbara

Santa Barbara, CA

- Developed the Broadband-plan Querying Tool (BQT), a large-scale web-scraping infrastructure that collected **1M+** residential broadband plans from 8 major ISPs, enabling the first nationwide comparative analysis of plan offerings.
- Created the Broadband Subscription Tier (BST) methodology, a hierarchical unsupervised learning framework that identifies user subscription tiers from crowdsourced measurements with **99% accuracy**.
- Built an NLP pipeline to classify **17,000** disaster-related tweets, enabling power and communication outage identification; evaluated **22 ML models** and achieved **90% accuracy**.

Awards

- ACM SIGCOMM Doctoral Dissertation Award, 2024.
- UCSB CS Department Outstanding Dissertation Award, 2023.
- UCSB CS Department Outstanding Publication Award, 2023.
- ACM SIGCOMM IMC Distinguished Paper Award, 2022.

Refereed Publications

- **Udit Paul**, Jiamo Liu, Vinothini Gunasekaran, Tejas N Narechania, Arpit Gupta, and Elizabeth Belding, “Decoding the Divide: Analyzing Disparities in Broadband Plans Offered by Major US ISPs”, Proceedings of the ACM SIGCOMM 2023 (**SIGCOMM’23**).
- **Udit Paul**, Jiamo Liu, Mengyang Gu, Arpit Gupta, and Elizabeth Belding, “The Importance of Contextualization of Crowdsourced Active Speed Test Measurements”, Proceedings of the ACM SIGCOMM Internet Measurement Conference 2022 (**IMC’22**). [*Best paper*].
- Tarun Mangla, **Udit Paul**, Arpit Gupta, Nicole Marwell, and Nick Feamster, “Internet Inequity in Chicago: Adoption, Affordability, and Availability”, Proceedings of the 50th Research Conference on Communications, Information and Internet Policy (**TPRC’22**).
- **Udit Paul**, Jiamo Liu, David Farias-Llerenas, Vivek Adarsh, Arpit Gupta, and Elizabeth Belding, “Characterizing Internet Access and Quality Inequities in California M-Lab Measurements”, Proceedings of the Conference on Computing and Sustainable Societies (**COMPASS’22**).
- Vivek Adarsh, Michael Nekrasov, **Udit Paul**, Tarun Mangla, Arpit Gupta, Morgan Vigil-Hayes, Ellen Zegura, and Elizabeth Belding, “Coverage is not binary: Quantifying mobile broadband quality in urban, rural, and tribal contexts”, Proceedings of the International Conference on Computer Communications and Networks (ICCCN) 2021 (**ICCN’21**).
- Vivek Adarsh, Michael Nekrasov, **Udit Paul**, Alex Ermakov, Arpit Gupta, Morgan Vigil-Hayes, Ellen Zegura, and Elizabeth Belding, “Too Late for Playback: Estimation of Video Stream Quality in Rural and Urban Contexts”, Proceedings of the Passive and Active Measurement Conference (**PAM’21**).
- Vivek Adarsh, Michael Nekrasov, **Udit Paul**, and Elizabeth Belding, “Estimation of congestion from cellular walled gardens using passive measurements”, IEEE Transactions on Mobile Computing, vol. 21, no. 10, 2021.
- **Udit Paul**, Alex Ermakov, Michael Nekrasov, Vivek Adarsh, and Elizabeth Belding, “#Outage: Detecting Power and Communication Outages from Social Networks”, Proceedings of the World Wide Web Conference (**WWW ’20**).
- Michael Nekrasov, Vivek Adarsh, **Udit Paul**, Esther Showalter, Ellen Zegura, Morgan Vigil-Hayes, and Elizabeth Belding, “Evaluating LTE coverage and quality from an unmanned aircraft system”, International Conference on Mobile Ad Hoc and Sensor Systems (**MASS’19**).

Technical Skills

Languages	Python, Ruby, C/C++, JavaScript, MySQL, PostgreSQL, Bash.
Developer Tools	Google Cloud Platform, Amazon Web Services, Android Studio.
Technologies/Frameworks	Linux, GitHub, Hadoop, Spark, Airflow, Docker, Tensorflow, Pytorch.